PRATICAL-8(a)

AIM:- **Perform data visualization using Python on any sales data.**

STEPS:-

Step 1:- **Install Required Libraries**

Before starting, install the necessary libraries.

Run the following command in your terminal or command prompt:

pip install pandas numpy matplotlib seaborn

Step 2:- Import Libraries

Now, open your Python script and import the required libraries:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

Step 3:- **Create Sample Sales Data**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate Sample Sales Data
dates = pd.date_range(start='2023-01-01', periods=100, freq='D')
categories = ['Electronics', 'Clothing', 'Groceries', 'Furniture']
data = {
    'Date': np.random.choice(dates, 200),
    'Category': np.random.choice(categories, 200),
    'Sales_Amount': np.random.randint(100, 1000, 200),
    'Units_Sold': np.random.randint(1, 20, 200)
}
df = pd.DataFrame(data)

# Convert Date column to datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Summary Statistics
print(df.describe())

# Sales Trend Over Time
plt.figure(figsize=(12, 6))
```

```
sns.lineplot(data=df.groupby('Date')['Sales_Amount'].sum().reset_index(), x='Date',
y='Sales_Amount')
plt.title('Sales Trend Over Time')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.show()

# Sales by Category
plt.figure(figsize=(10, 5))
sns.barplot(data=df.groupby('Category', as_index=False)['Sales_Amount'].sum(), x='Category',
y='Sales_Amount', palette='viridis')
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Sales Amount')
plt.show()

# Distribution of Sales Amounts
plt.figure(figsize=(10, 5))
sns.histplot(df['Sales_Amount'], bins=20, kde=True, color='blue')
plt.title('Distribution of Sales Amounts')
plt.xlabel('Sales Amount')
plt.ylabel('Frequency')
plt.show()

# Scatter Plot: Sales Amount vs. Units Sold
plt.figure(figsize=(10, 5))
sns.scatterplot(data=df, x='Units_Sold', y='Sales_Amount', hue='Category',
palette='coolwarm')
plt.title('Sales Amount vs. Units Sold')
plt.xlabel('Units Sold')
plt.ylabel('Sales Amount')
plt.show()
```

data_analysis.py - C:/Users/Saurav/OneDrive/文件/data_analysis.py (3.12.9)

File Edit Format Run Options Window Help

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate Sample Sales Data
dates = pd.date_range(start='2023-01-01', periods=100, freq='D')
categories = ['Electronics', 'Clothing', 'Groceries', 'Furniture']
data = {
    'Date': np.random.choice(dates, 200),
    'Category': np.random.choice(categories, 200),
    'Sales_Amount': np.random.randint(100, 1000, 200),
    'Units_Sold': np.random.randint(1, 20, 200)
}
df = pd.DataFrame(data)

# Convert Date column to datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Summary Statistics
print(df.describe())

# Sales Trend Over Time
plt.figure(figsize=(12, 6))
sns.lineplot(data=df.groupby('Date')['Sales_Amount'].sum().reset_index(), x='Date', y='Sales_Amount')
plt.title('Sales Trend Over Time')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.show()

# Sales by Category
plt.figure(figsize=(10, 5))
sns.barplot(data=df.groupby('Category', as_index=False)['Sales_Amount'].sum(), x='Category', y='Sales_Amount', palette='viridis')
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Sales Amount')
plt.show()

# Distribution of Sales Amounts
plt.figure(figsize=(10, 5))
sns.histplot(df['Sales_Amount'], bins=20, kde=True, color='blue')
plt.title('Distribution of Sales Amounts')
plt.xlabel('Sales Amount')
plt.ylabel('Frequency')
plt.show()
```

Output:-

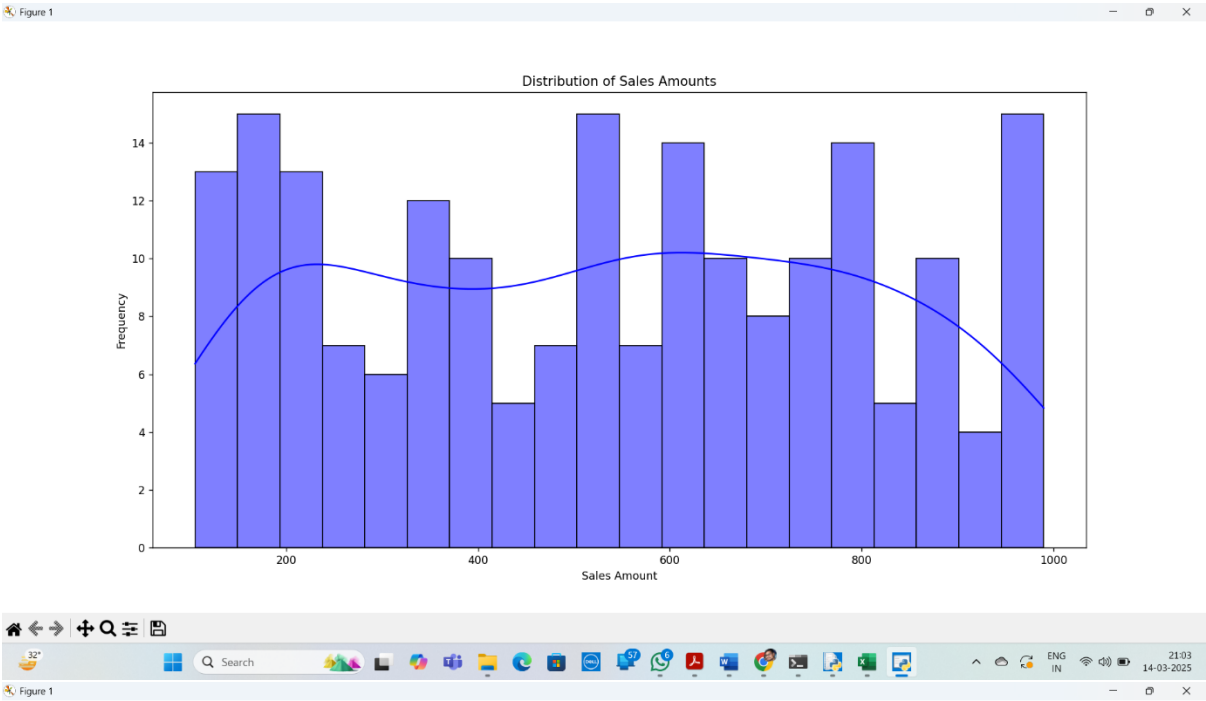


Figure 1
Distribution of Sales Amounts
Frequency
Sales Amount
0
2
4
6
8
10
12
14
200
400
600
800
1000
Search
ENG
IN
21:03
14-03-2025

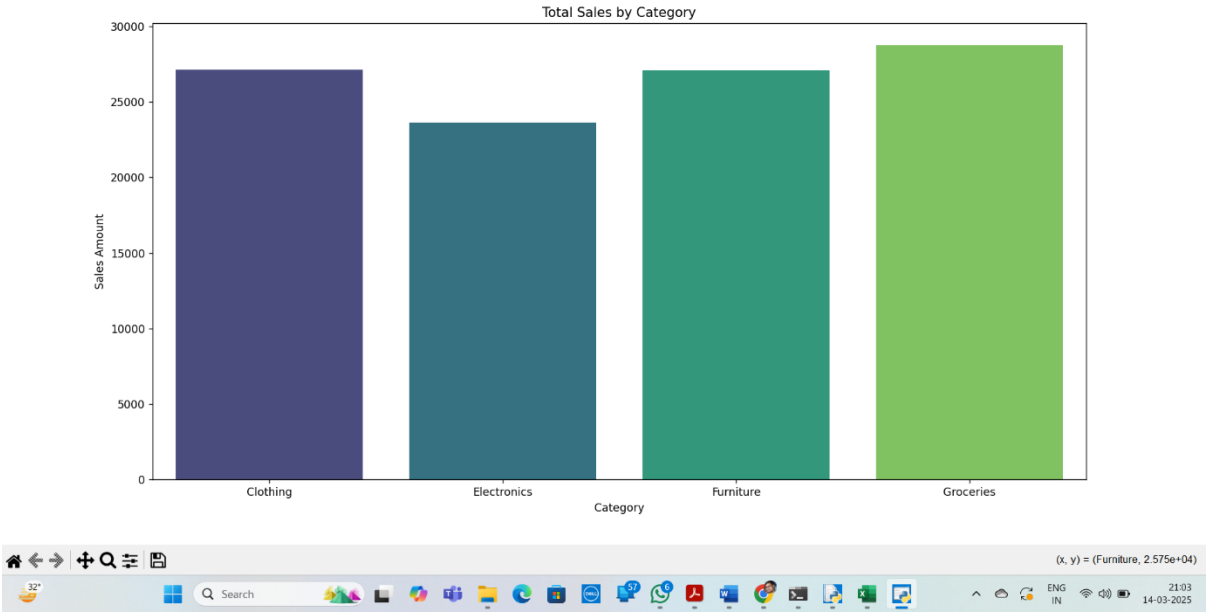


Figure 1
Total Sales by Category
Sales Amount
Category
0
5000
10000
15000
20000
25000
30000
Clothing
Electronics
Furniture
Groceries
(x, y) = (Furniture, 2.575e+04)
Search
ENG
IN
21:03
14-03-2025

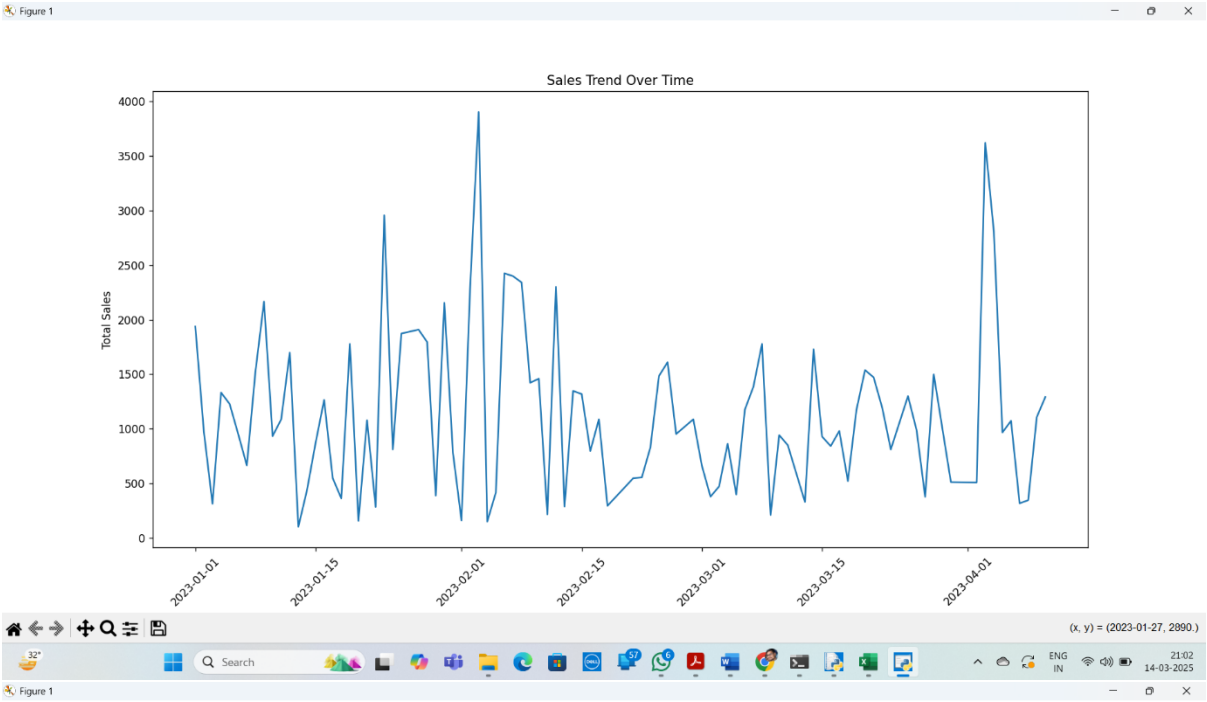


Figure 1
Sales Trend Over Time
Total Sales
4000
3500
3000
2500
2000
1500
1000
500
0
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01
(x, y) = (2023-01-27, 2890.)

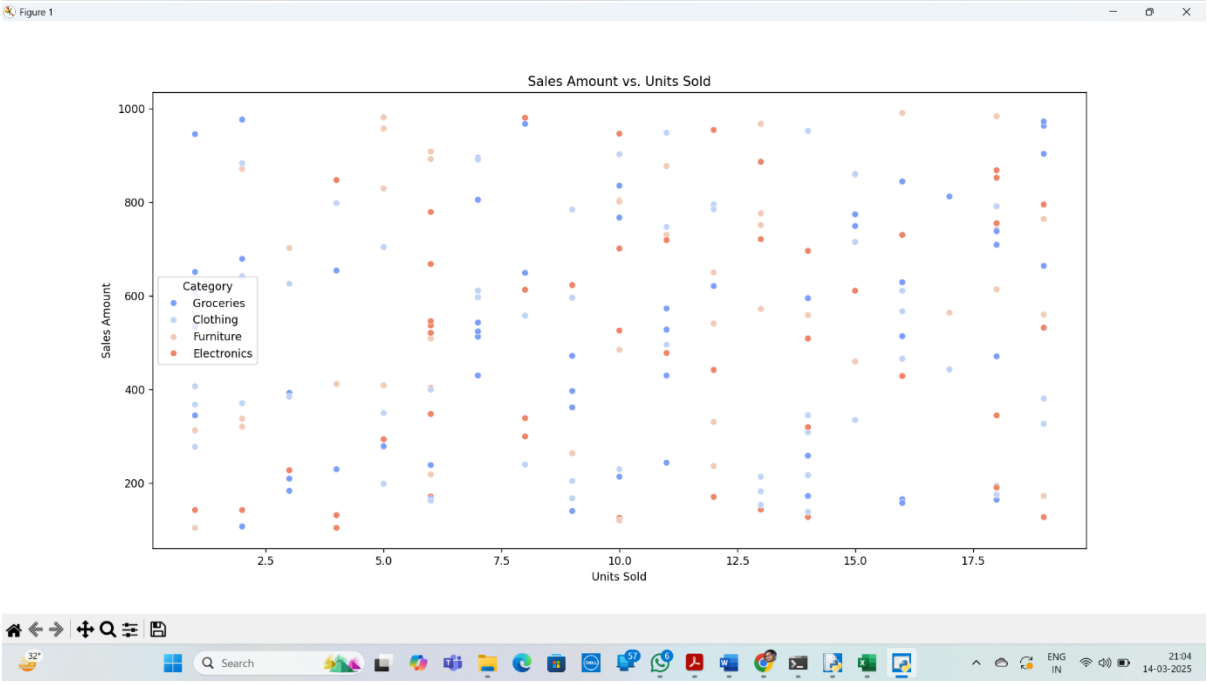


Figure 1
Sales Amount vs. Units Sold
Sales Amount
1000
800
600
400
200
Category
Groceries
Clothing
Furniture
Electronics
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Units Sold

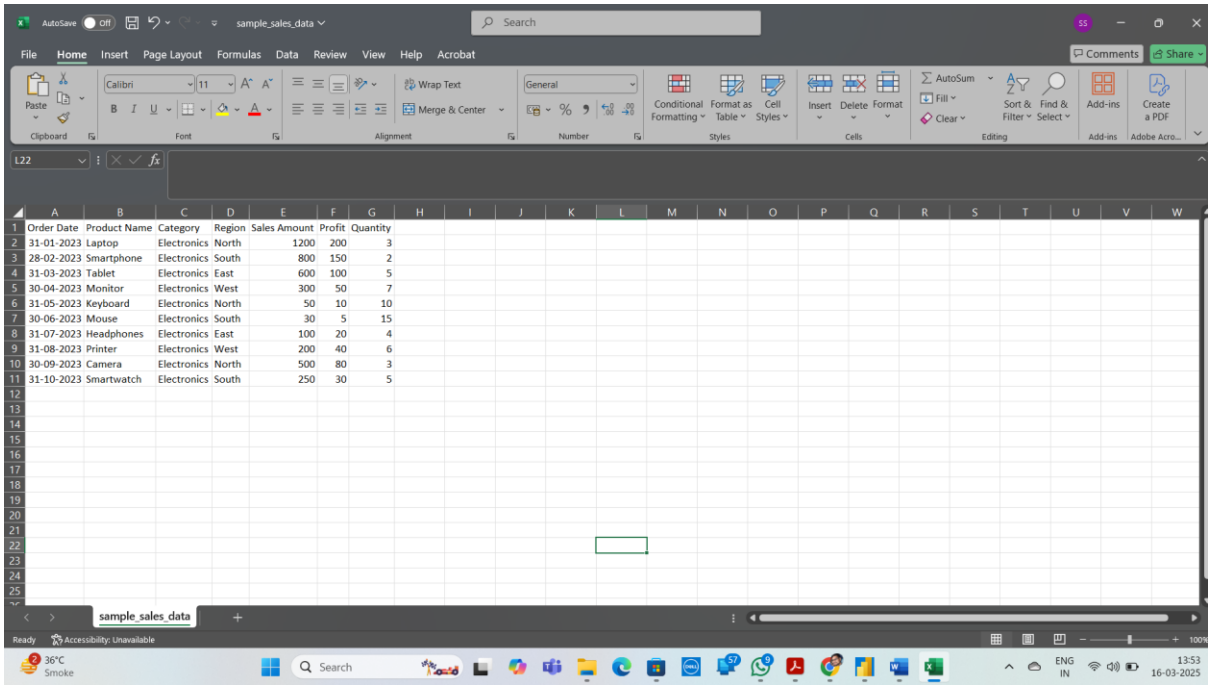
PRATICAL-8(b)

AIM:- Perform data visualization using PowerBI on any sales data

Steps:-

Step 1:- Download & Install Power BI

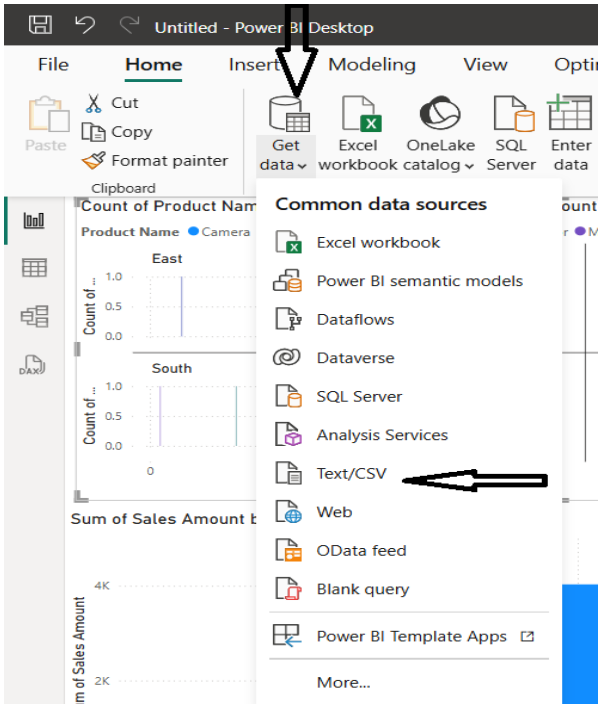
Step 2:- create a sample data



Order Date	Product Name	Category	Region	Sales Amount	Profit	Quantity
31-01-2023	Laptop	Electronics	North	1200	200	3
28-02-2023	Smartphone	Electronics	South	800	150	2
31-03-2023	Tablet	Electronics	East	600	100	5
30-04-2023	Monitor	Electronics	West	300	50	7
31-05-2023	Keyboard	Electronics	North	50	10	10
30-06-2023	Mouse	Electronics	South	30	5	15
31-07-2023	Headphones	Electronics	East	100	20	4
31-08-2023	Printer	Electronics	West	200	40	6
30-09-2023	Camera	Electronics	North	500	80	3
31-10-2023	Smartwatch	Electronics	South	250	30	5

Step 3 :- Import the Sales Dataset

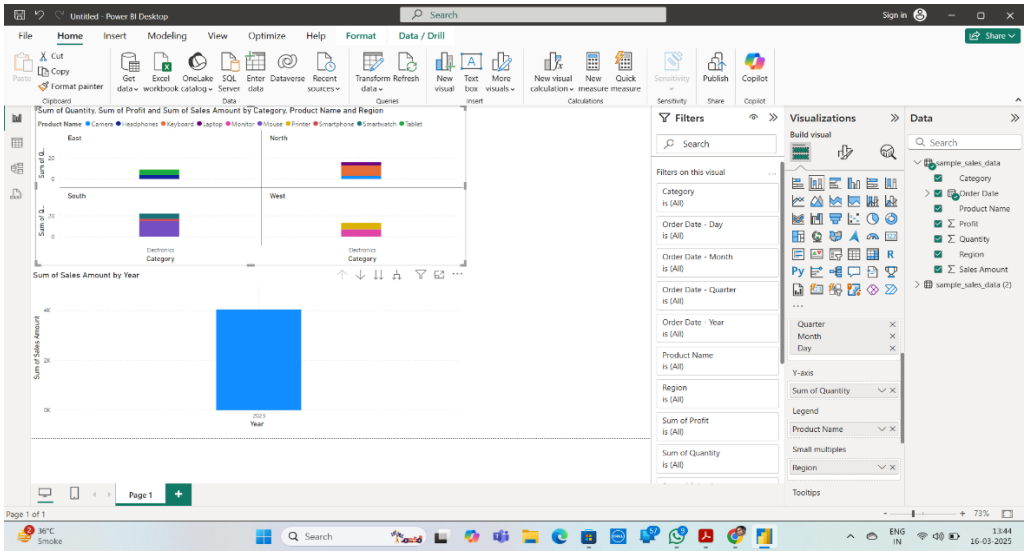
- Click "Home" > "Get Data".
- Choose your data source:
  - Excel (XLSX/CSV)
  - SQL Server
  - Online Services (Google Sheets, SharePoint, etc.)
- Browse and select the dataset.
- Click "Load" to import.



Step 4 :- Clean & Transform Data (Power Query)

Step 5 :- Create Data Visualizations

- Click on **"Line Chart"** in the "Visualizations" pane.
- Drag **Order Date** to the **X-axis**.
- Drag **Sales Amount** to the **Y-axis**.
- Customize the chart (format labels, add title, etc.).

Step 6:- Add Filters & Interactivity

Output:-